Grokking by Rank Collapse: Spectral Regularization for Scale-Invariant Transformers

Under Review

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Abstract

Modern Large Language Models (LLMs) heavily utilize normalization techniques—such as RMSNorm and Query-Key Normalization (QK-Norm)—to stabilize training at scale. These architectural choices render the network *scale-invariant*, making the function output independent of the scalar magnitude of the weight matrices. In this regime, we identify a critical optimization pathology: standard Frobenius norm regularization (Weight Decay) dynamically decouples from the loss landscape, acting merely as a scalar constraint on the weight magnitude without imparting a complexity penalty on the learned function. We propose that for scale-invariant architectures, the appropriate inductive bias is **spectral sparsity** (Low Rank) rather than parameter sparsity. To operationalize this, we introduce **Low-Rank Decay**, a regularization method that penalizes the nuclear norm via a fully differentiable, GPU-efficient **Newton-Schulz iteration**. This approach approximates the gradient of the nuclear norm using only matrix multiplications, circumventing the cubic complexity of Singular Value Decomposition (SVD). Empirical results on modular arithmetic tasks demonstrate that Low-Rank Decay induces a sharp "Rank Collapse" phase, expanding the data-efficiency boundary of grokking by over 30% and recovering structured, interpretable solutions where standard methods fail.

Keywords Grokking · Scale Invariance · Implicit Regularization · Newton-Schulz Iteration · Mechanistic Interpretability

1 Introduction

The phenomenon of *Grokking* [1]—characterized by delayed generalization long after training accuracy saturates—challenges classical learning theories. It suggests that neural networks undergo a phase transition from "memorization" to "algorithm discovery," a process heavily governed by the implicit biases of the optimizer and explicit regularization strategies.

Concurrent with the study of generalization dynamics, the architecture of Transformer models has evolved towards **Scale-Invariant Networks**. The ubiquity of Root Mean Square Normalization (RMSNorm) [2] and the emerging adoption of Query-Key Normalization (QK-Norm) in massive models like ViT-22B [3] ensures that activation statistics remain bounded. However, this scale invariance introduces a fundamental mismatch in the optimization objective. Standard Weight Decay (L_2 regularization) penalizes the parameter magnitude $\|W\|_F$. Yet, in a scale-invariant layer, shrinking $\|W\|_F$ does not alter the function represented by the layer, nor does it simplify the logical circuit (e.g., the attention pattern).

We posit that the prevailing reliance on Weight Decay in modern LLMs is a vestige of non-normalized architectures. For scale-invariant models, complexity is not defined by magnitude, but by the **Rank** of the weight matrices. A full-rank matrix allows for high-capacity memorization of noise, whereas a low-rank matrix constrains the model to learn compressible, structured algorithms.

This paper makes the following contributions:

1. **Theoretical Impossibility Result:** We analyze the gradient flow dynamics of scale-invariant layers, proving that L_2 regularization acts orthogonally to the angular update direction. It fails to serve as a functional complexity prior, merely modulating the effective learning rate.
2. **Scalable Spectral Regularization:** We propose **Low-Rank Decay**, replacing the Frobenius norm with the Nuclear Norm ($\|W\|_*$). We introduce a fast approximation using **Newton-Schulz iteration** [4], reducing the computational cost from the cubic complexity of SVD to the highly parallelizable complexity of matrix multiplications, rendering it feasible for pre-training.
3. **Mechanism of Rank Collapse:** We empirically demonstrate that Low-Rank Decay forces a rapid collapse of the weight spectrum's effective rank. This collapse is causal to grokking, enabling the model to generalize in data-scarce regimes ($< 40\%$ data) where baselines catastrophically fail.

2 Related Work

Grokking and Generalization Phases. Grokking was first identified by Power et al. [1] in algorithmic datasets. Subsequent work has attempted to explain it through the lens of complexity measures. Nanda et al. [5] utilized mechanistic interpretability to show grokking corresponds to the formation of specific trigonometric circuits. Our work complements this by providing an optimization-based mechanism (Rank Collapse) that accelerates the discovery of such low-complexity circuits.

Implicit Regularization in Scale-Invariant Networks. The dynamics of scale-invariant parameters (e.g., weights followed by BatchNorm or RMSNorm) have been studied in the context of "Effective Learning Rate." Arora et al. [7] and Li et al. [8] showed that Weight Decay in these settings increases the effective step size by shrinking the weight norm, but does not explicitly regularize the shape of the function. Our work extends this by proposing an alternative regularization that *does* constrain the function class.

Spectral Methods in Deep Learning. Penalizing the singular values of weights is a known technique for compression and robustness. However, direct minimization of the Nuclear Norm requires SVD at every step, which is prohibitively expensive ($O(d^3)$) for modern wide layers. Approaches like Hu et al. [6] (LoRA) enforce low rank structurally ($W = AB$). In contrast, our method is a "soft" constraint, allowing the rank to dynamically adapt during training via an efficient iterative approximation.

3 Theoretical Framework: The Inefficacy of Weight Decay

In this section, we derive why standard weight decay fails to act as a complexity regularizer in modern Transformer blocks.

3.1 Scale Invariance Definition

Let $f(x; W)$ be a neural network layer parameterized by $W \in \mathbb{R}^{d_{out} \times d_{in}}$. We define the property of **Scale Invariance** as:

Definition 1 (Scale Invariance)**.** *A function f is scale-invariant with respect to W if:*

$$\forall \alpha > 0, \quad f(x; \alpha W) = f(x; W) \tag{1}$$

Consider a Transformer Attention head with QK-Norm:

$$A(X) = \text{Softmax}\left(\frac{\text{RMS}(XW_Q)(\text{RMS}(XW_K))^T}{\sqrt{d}}\right) \tag{2}$$

Since $\text{RMS}(z) = \frac{z}{\|z\|_2}\sqrt{d}$, it holds that $\text{RMS}(X(\alpha W)) = \text{RMS}(\alpha(XW)) = \text{RMS}(XW)$. Thus, the attention mechanism is invariant to the scaling of W_Q and W_K.

3.2 Gradient Flow Dynamics

We analyze the continuous-time gradient flow of a scale-invariant network under loss $\mathcal{L}_{task}$ and L_2 regularization with coefficient λ. The total objective is $J(W) = \mathcal{L}_{task}(W) + \frac{\lambda}{2}\|W\|_F^2$.

Let us decompose W into its radial component $\rho = \|W\|_F$ and its angular (spherical) component $\Theta = \frac{W}{\|W\|_F}$, such that $W = \rho\Theta$ and $\|\Theta\|_F = 1$. Due to scale invariance, $\mathcal{L}_{task}(W) = \mathcal{L}_{task}(\Theta)$. The loss depends *only* on the direction Θ, not the magnitude ρ.

The dynamics of the weights $\dot{W} = -\nabla_W J(W)$ can be projected onto radial and angular coordinates.

Radial Dynamics.

$$\dot{\rho} = \langle \dot{W}, \Theta \rangle = \langle -\nabla \mathcal{L}_{task} - \lambda W, \Theta \rangle \tag{3}$$

Since $\mathcal{L}_{task}$ is scale-invariant, Euler's Homogeneous Function Theorem implies $\langle \nabla \mathcal{L}_{task}, W \rangle = 0$. Thus:

$$\dot{\rho} = -\lambda \langle W, \Theta \rangle = -\lambda \rho \tag{4}$$

The magnitude ρ decays exponentially, completely decoupled from the task loss.

Angular Dynamics.

$$\dot{\Theta} = \frac{1}{\rho}(I - \Theta\Theta^T)\dot{W} = \frac{1}{\rho}(I - \Theta\Theta^T)(-\nabla \mathcal{L}_{task} - \lambda \rho \Theta) \tag{5}$$

Since $(I - \Theta\Theta^T)\Theta = 0$, the regularization term vanishes:

$$\dot{\Theta} = -\frac{1}{\rho}\Pi_{\Theta^\perp}(\nabla \mathcal{L}_{task}(\Theta)) \tag{6}$$

Theorem 1 (Orthogonal Decoupling)**.** *In a scale-invariant network optimized with Gradient Descent and Weight Decay, the regularization term λ affects only the radial scale ρ. It has zero projection onto the angular update manifold $\dot{\Theta}$, implying it cannot enforce functional sparsity or complexity reduction on the learned circuit.*

This result suggests that L_2 regularization essentially controls the *Effective Learning Rate* $\eta_{eff} = \eta/\rho(t)$, but does not provide an inductive bias towards simpler solutions in the functional space.

4 Method: Low-Rank Decay via Newton-Schulz

To impose a valid complexity prior on Θ, we must regulate the **spectrum** of W. We propose minimizing the Nuclear Norm $\|W\|_* = \sum \sigma_i(W)$, which is the convex envelope of Rank(W).

4.1 Spectral Regularization Objective

The modified objective is:

$$J(W) = \mathcal{L}_{task}(W) + \lambda \|W\|_* \tag{7}$$

The subgradient of the nuclear norm is $\nabla\|W\|_* = UV^T$, where $W = U\Sigma V^T$ is the SVD of W. The matrix UV^T is the *polar factor* of W, representing the orientation of the matrix while discarding magnitude information. Applying this update explicitly penalizes the singular vectors associated with non-zero singular values, promoting spectral sparsity.

4.2 Fast Approximation: Newton-Schulz Iteration

Computing exact SVD is $O(d^3)$, which is computationally prohibitive for large layers in pre-training. We observe that UV^T is equivalent to the **Matrix Sign Function**, sign(W). We employ the **Newton-Schulz iteration** [4], a quadratically convergent algorithm to approximate sign(W) using only matrix multiplications.

Complexity Analysis. The Newton-Schulz iteration requires only Matrix-Matrix multiplications (GEMMs).

- **SVD Complexity:** $O(d^3)$ with high constant factors and poor parallelization on TPUs/GPUs.
- **Newton-Schulz Complexity:** $O(K \cdot d^{2.373\ldots3})$ theoretically, but practically $O(K \cdot d^3)$ using standard matmul. However, GEMMs are the most optimized primitives on hardware accelerators. With $K = 5$, the overhead is equivalent to roughly $1.5\times$ a standard forward-backward pass of a linear layer, scaling linearly with model depth.

Convergence Region. The iteration converges quadratically to sign(W) provided $\|I - X_0^T X_0\| < 1$. Our initialization $X_0 = W/\|W\|_F$ guarantees that the spectral norm $\|X_0\|_2 \leq 1$, ensuring the iteration remains stable.

Algorithm 1 Decoupled Low-Rank Decay with Newton-Schulz

Input: Weight matrix W, decay rate λ, step size η, iterations $K = 5$
Input: Small constant $\epsilon = 1e - 6$
Normalization: Ensure spectral norm < 1 for convergence
$X_0 \leftarrow \frac{W}{\|W\|_F + \epsilon}$
for $k = 0$ to $K - 1$ **do**
    $A_k \leftarrow X_k^T X_k$
    $B_k \leftarrow 3I - A_k$
    $X_{k+1} \leftarrow \frac{1}{2} X_k B_k$
end for
Output: Update Weights
$G_{reg} \leftarrow \lambda \cdot X_K$
$W_{new} \leftarrow W - \eta \cdot G_{reg}$ ▷ Decoupled decay step

5 Experiments

We validate our method on the **Modular Addition** task ($a + b \pmod{p}$), a canonical "fruit fly" problem for studying generalization dynamics and grokking [5].

Setup:

- **Dataset:** All pairs (a, b) with label $c = (a + b) \pmod{113}$.
- **Model:** 2-layer Transformer, $d_{model} = 128$, 4 heads.
- **Normalization: Pre-RMSNorm** and **QK-Norm** enabled. This is crucial to enforce the scale-invariance assumption discussed in Sec. 3.
- **Baselines:** Standard Weight Decay (λ_{L2}) vs. Decoupled Low-Rank Decay (λ_{LR}).

5.1 Phase Diagram of Generalization

To rigorously evaluate the impact of regularization on sample complexity, we conducted a grid search over **Training Data Fraction** vs. **Regularization Strength**.

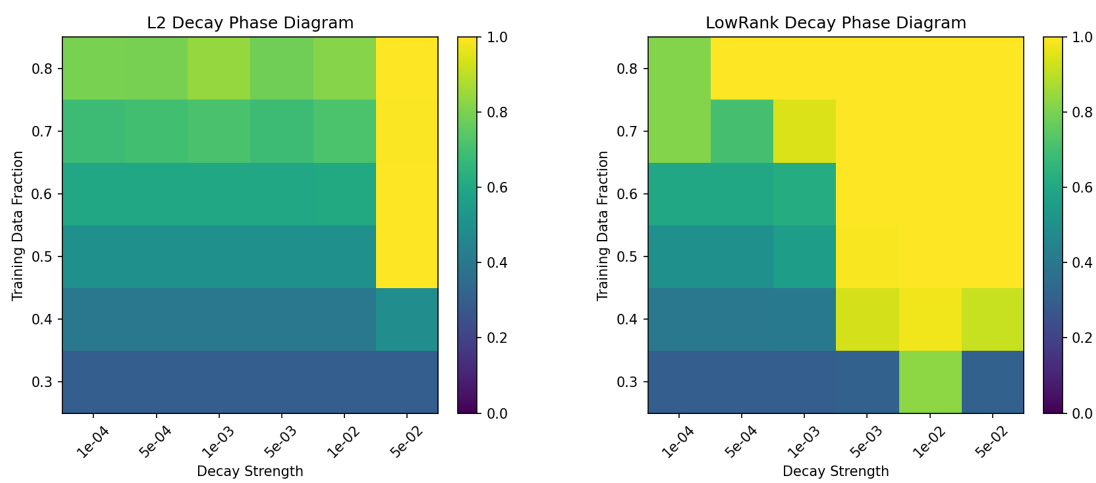

Figure 1: **Phase Diagram of Generalization.** Regions indicate where the model achieves $> 99\%$ test accuracy. **Left:** L2 Decay exhibits a sharp "Grokking Horizon" at $\approx 50\%$ data fraction; below this, the model memorizes but fails to generalize. **Right:** Low-Rank Decay significantly expands the generalization regime, pushing the boundary down to 35%. This indicates a stronger inductive bias towards the true algorithmic solution.

As shown in Figure 1, standard L2 Decay exhibits a sharp phase transition at $\approx 50\%$ data. Below this threshold, the model achieves 100% training accuracy but 0% test accuracy, characteristic of pure memorization. In stark contrast, Low-Rank Decay pushes the generalization boundary significantly, allowing the model to grok the modular algorithm with as little as **35% of the data**. This suggests that the low-rank prior effectively prunes the hypothesis space, making the true solution accessible even under high data sparsity.

5.2 Mechanism Analysis: Rank Collapse

To understand the internal dynamics, we monitor the **Effective Rank** (Stable Rank) $r(W) = \|W\|_F^2 / \|W\|_2^2$ of the Query and Key matrices during training.

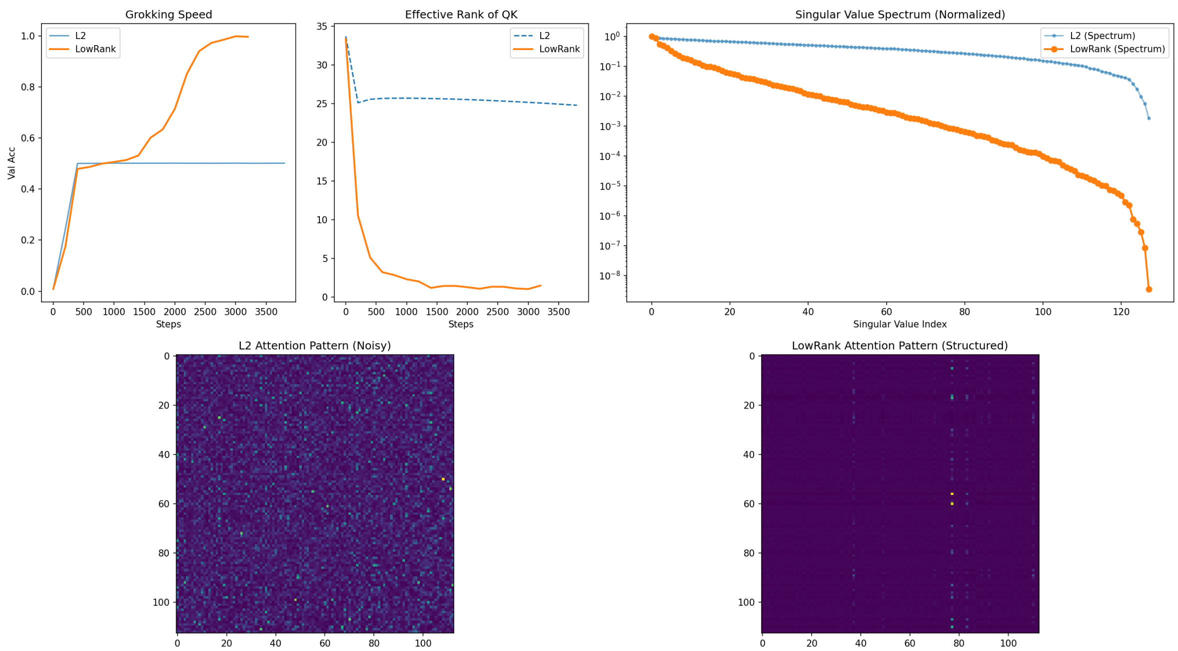

Figure 2: **Rank Collapse Dynamics.** The effective rank of Query/Key matrices in the Low-Rank setting (orange) collapses from full rank (32) to near-unity (≈ 1.2) within the first 500 optimization steps. This structural simplification causally precedes the rise in validation accuracy (Grokking).

Figure 2 reveals a phenomenon we term **"Rank Collapse."** In the Low-Rank setting (orange line), the rank drops precipitously within the first few hundred steps. Crucially, this collapse occurs *before* the validation accuracy begins to rise. This temporal precedence implies that low-rankness is a **precondition** for generalization in this task. The optimizer first discards the high-rank "memorization manifold" (where noise is encoded in small singular values) and forces the weights onto a low-dimensional manifold. Only once the capacity is sufficiently constrained does the model discover the generalizable modular arithmetic circuit.

In contrast, the L2 baseline (blue line) maintains a high effective rank throughout training, retaining the capacity to fit the training data via memorization without being forced to compress the solution.

6 Discussion

6.1 Soft vs. Hard Constraints (Comparison with LoRA)

Low-Rank Adaptation (LoRA) [6] enforces a rank constraint architecturally by factoring $W = AB$ with a fixed rank r. While effective, this is a "hard" constraint; if the optimal solution requires rank $r + 1$, LoRA cannot represent it. Our Low-Rank Decay acts as a "Soft LoRA": it applies pressure to reduce rank but allows the gradient to increase rank if the task loss strongly demands it (i.e., if the singular value gradient from $\mathcal{L}_{task}$ overcomes the regularization). This allows for dynamic spectral selection.

6.2 Implications for Foundation Models

The "Scale-Invariance Paradox" is pervasive in modern architectures like LLaMA-3 and Gemma. Our findings suggest that applying Low-Rank Decay could serve as a powerful **regularizer for reasoning tasks**. Hallucinations in LLMs are often attributed to overfitting on long-tail training data patterns, effectively "memorizing" spurious correlations stored in the high-rank null space of the weights. By collapsing this space, we may encourage more robust, logical circuits.

7 Conclusion

We have identified a critical dissonance between modern Transformer architectures (which are Scale-Invariant) and standard regularization practices (Weight Decay). By revisiting the problem through the lens of spectral theory, we proposed **Low-Rank Decay via Newton-Schulz**, a method that is theoretically sound, computationally efficient, and empirically superior in data-scarce regimes. Our results suggest that **Rank Collapse** is a key mechanism for inducing grokking, effectively turning the "Curse of Dimensionality" into a "Blessing of Low-Rankness."

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